

Bias Geometry Under Matryoshka Representation Learning: A Two-Phase Study from Word Embeddings to Sentence Transformers

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Abstract—Matryoshka Representation Learning (MRL) [1] has rapidly become the dominant paradigm for adaptive-length text embeddings in production AI systems. Its nested loss formulation enables any prefix of the output vector to serve as a valid representation, making it the training objective of choice for models deployed by OpenAI (text-embedding-3 series) [2], Google Gemini Embeddings [3], Nomic [4], and Alibaba’s Qwen3 embedding series [5]—collectively serving hundreds of millions of API calls per day. In practice, engineers routinely truncate these embeddings from their full dimensionality (768–2560) down to 64–256 dimensions to reduce memory footprint and accelerate nearest-neighbour search.

A critical and heretofore unstudied consequence of this practice is its effect on social bias. Embedding models are known to encode gender, racial, and occupational stereotypes inherited from their training corpora [9], [10], yet all prior bias evaluation work operates at a single fixed full dimension. No existing study has asked what the MRL training objective does to bias geometry when embeddings are truncated.

We present the *first systematic two-phase study* of this question. Phase 1 establishes causal ground truth through a fully controlled experiment: we train Standard Skip-Gram with Negative Sampling (SGNS) and our MRL variant from scratch on an *identical* 50M-token corpus with identical vocabulary, isolating the effect of the Relational Distillation objective on bias geometry across six prefix dimensions $d \in \{50, 100, 150, 200, 250, 300\}$. Phase 2 extends the inquiry to six diverse production transformer models (MPNet Standard/MRL, Nomic-embed-v1.5, mxbai-embed-large, Qwen3-0.6B, Qwen3-4B), evaluating both intrinsic bias geometry and downstream behavioural bias.

Across both phases, we find that MRL does *not* debias embeddings; instead it *reorganises* bias from local neighbourhood structure to global directional structure and *front-loads* bias energy into early prefix dimensions. Critically, intrinsic and downstream bias metrics systematically decouple across all six transformer models—a warning that standard full-dimension bias audits are insufficient for MRL deployments. We introduce the *Anisotropy Factor* $\alpha(d)$ as a novel diagnostic that consistently distinguishes MRL from Standard training.

Index Terms—Matryoshka Representation Learning, word embeddings, social bias, gender bias, sentence transformers, dimensionality reduction, WEAT, fairness, bias geometry

I. INTRODUCTION

Matryoshka Representation Learning (MRL), introduced by Kusupati et al. [1], modifies the embedding training objective

so that any prefix $\mathbf{e}_{1:d}$ of the full output vector $\mathbf{e}_{1:D}$ constitutes a meaningful representation at that lower dimensionality. This property has proven enormously valuable for production deployment: rather than training and maintaining separate models for different computational budgets, a single MRL model can serve all use cases at inference time through prefix truncation.

The practical adoption of MRL has been swift and widespread. OpenAI’s text-embedding-3 series explicitly supports dimension truncation via MRL [2]. Google’s Gecko embeddings [3] and the Gemini Embedding API adopt the same paradigm. The Sentence-Transformers library natively provides a MatryoshkaLoss class [6], making it accessible to any practitioner. Among open models, Nomic-embed-text-v1.5 [4], mxbai-embed-large, and the Qwen3 embedding series (Alibaba, 2024) all employ MRL as their primary training objective. Recent benchmarks on the Massive Text Embedding Benchmark (MTEB) demonstrate that MRL-trained models retain approximately 98% of their full-dimensional performance even when truncated to 8–10% of their original size [7].

In retrieval-augmented generation (RAG) pipelines, semantic search engines, and recommendation systems, the practical benefit translates to engineers truncating MRL embeddings from their full dimensionality to 64 or 128 dimensions—achieving 5–8 $\times$ memory reduction with minimal semantic accuracy loss [1].

The unexamined assumption. Embedding models are well-documented sources of social bias. Bolukbasi et al. [9] showed that Word2Vec encodes occupational gender stereotypes; Caliskan et al. [10] demonstrated that these biases replicate human implicit associations. An extensive subsequent literature has studied, measured, and proposed mitigations for these biases [11]–[14]. However, *all* of this work evaluates bias at a single, fixed, full dimension. The MRL truncation that is ubiquitous in deployment is simply not studied from a bias perspective.

This creates a dangerous gap. If MRL front-loads bias into early prefix dimensions, a model that passes a bias audit

at $d = 768$ may exhibit dramatically amplified bias signals at the $d = 64$ configuration actually used in production. Conversely, if MRL disperses bias into later dimensions, standard full-dim audits give falsely reassuring results for truncated deployments. Either scenario renders the status quo of bias evaluation insufficient.

Why two phases? Establishing causal claims about the MRL training objective requires controlled conditions—identical data, identical vocabulary, only the loss function varying. Word2Vec SGNS affords exactly this control at the cost of static, word-level embeddings and modest dimensionality. Study 1 exploits this to establish causal ground truth. Study 2 then asks whether the identified geometric phenomena generalise to the contextualised, high-dimensional transformer models that dominate modern MRL deployments. Together, the two phases provide both causal rigour and real-world generalisability.

Contributions

- 1) **First controlled study of bias geometry under MRL training.** We train Standard SGNS and MRL from scratch on an identical corpus, isolating the causal effect of Relational Distillation on bias across ten geometric and seven standard bias metrics, supplemented by fifteen deep-dive experiments.
- 2) **Characterisation of bias reorganisation.** We show that MRL transforms *how* bias is encoded without changing *how much* total bias exists: bias moves from local neighbourhood composition to global angular structure, is front-loaded in prefix dimensions, and is redistributed across words in a way uncorrelated with the Standard model’s profile ($r = -0.12$, $p = 0.45$).
- 3) **Metric decoupling across six transformer models.** We demonstrate that intrinsic bias metrics (WEAT, $C(d)$) systematically decouple from downstream preference metrics (StereoSet, CrowS-Pairs), and introduce the Anisotropy Factor $\alpha(d)$ as a novel diagnostic that consistently distinguishes MRL from Standard training.

II. BACKGROUND AND RELATED WORK

A. Matryoshka Representation Learning

Standard embedding models produce a single fixed-dimensional vector. MRL modifies the training objective so that every prefix of the output is independently useful [1]. This is achieved by computing the standard task loss at multiple prefix dimensions simultaneously and summing the contributions:

$$\mathcal{L}_{\text{MRL}} = \sum_{d \in M} w_d \cdot \mathcal{L}^{(d)} \quad (1)$$

where M is the set of target prefix dimensions and w_d are importance weights. The key insight is that the early dimensions of the embedding must carry enough information to be useful independently, not merely as prefixes of a larger vector.

Relational Distillation (“MRL”) addresses magnitude exploitation by applying the NS loss only at the full dimension D and adding a distillation auxiliary loss for each prefix:

$$\mathcal{L}_{\text{MRL}} = \mathcal{L}_{\text{NS}}^{(D)} + \frac{1}{|M| - 1} \sum_{d < D} \left[\alpha \cdot \text{KL}(P_T \| P_S^{(d)}) + \beta \cdot \text{MSE}(\cos_d, \cos_D) \right] \quad (2)$$

For each prefix d , a teacher module computes a $B \times B$ cosine similarity matrix from full-dimensional embeddings (via a 2-layer MLP projection head, L2-normalised, temperature-scaled). The student prefix must match this relational structure. Teacher gradients are stopped (`.detach()`), so only prefix dimensions and their projection heads are updated via distillation; full-dimensional embeddings train purely via NS loss. Projection heads are discarded at inference.

This Relational Distillation mechanism is the basis for the widespread MRL adoption described above: by distilling full-dimensional similarity structure into each prefix, the model learns to encode salient relational features into early dimensions. As we demonstrate in this paper, gender and occupational bias patterns are among those salient features.

B. Social Bias in Word and Sentence Embeddings

The study of social bias in embeddings has progressed through several waves of research. Bolukbasi et al. [9] established that Word2Vec encodes gender stereotypes with the analogy “man : computer programmer :: woman : homemaker” and proposed hard debiasing via gender subspace projection. Caliskan et al. [10] introduced the Word Embedding Association Test (WEAT) and showed that embedding associations replicate implicit associations from the psychology literature. Garg et al. [14] demonstrated that embeddings quantify historical shifts in gender and ethnic stereotypes.

A second wave critiqued debiasing efficacy. Gonen and Goldberg [11] showed that Bolukbasi’s hard debiasing merely conceals bias rather than removing it. Ethayarajh et al. [13] introduced RIPA (Relational Inner Product Analysis) to understand the geometric nature of biased associations. Dev and Phillips [12] proposed the Embedding Coherence Test (ECT) to measure the symmetry of occupation-gender associations. Mimno and Thompson [15] characterised the geometric properties of SGNS, specifically the narrow-cone effect where all embedding vectors cluster near the mean direction.

For sentence encoders, May et al. [18] extended WEAT to contextualised sentence representations. Nadeem et al. [16] introduced StereoSet for measuring stereotypical bias in language models, and Nangia et al. [17] provided the CrowS-Pairs dataset for preference-based bias measurement. De-Arteaga et al. [19] introduced Bias in Bios, which measures downstream bias through occupation classification with gendered biographies.

The dimension-dependence gap. All of the above work evaluates embeddings at a single, fixed, full dimension. No prior study systematically varies prefix dimension and measures how bias evolves under standard versus MRL training. This is the gap our work fills.

III. STUDY 1: CONTROLLED WORD EMBEDDING EXPERIMENT

A. Motivation for the Word-Level Phase

Before studying bias in complex production transformers, we require a setting where the *only* variable is the training objective. Word2Vec Skip-Gram with Negative Sampling (SGNS) provides this. Two models trained on identical data with identical vocabulary differ only in their loss function; any observed difference in bias geometry is unambiguously attributable to the MRL objective. This causal clarity is not possible in transformer settings where architecture, pre-training data, and fine-tuning data all differ.

B. Embedding Models

We train two models from scratch on identical data.

1) *Standard SGNS (Baseline)*: The standard Skip-Gram objective [8] maximises the log probability of context words given a centre word. With negative sampling, the loss for a single (centre, context) pair (c, w) is:

$$\mathcal{L}_{\text{NS}} = \log \sigma(\mathbf{u}_w^\top \mathbf{v}_c) + \sum_{k=1}^K \mathbb{E}_{w_k \sim P_n} [\log \sigma(-\mathbf{u}_{w_k}^\top \mathbf{v}_c)] \quad (3)$$

where $\mathbf{v}_c \in \mathbb{R}^D$ is the centre word embedding, $\mathbf{u}_w \in \mathbb{R}^D$ is the context word embedding, σ is the sigmoid function, $K = 10$ negatives are drawn per positive pair, and $P_n(w) \propto f(w)^{3/4}$ is the subsampled noise distribution (frequency raised to the 3/4 power, following [8]). Window size is 5. Embedding dimension $D = 300$.

The optimisation proceeds via stochastic gradient descent with a linearly decaying learning rate. Centre embeddings $\mathbf{v}_c$ are the final word representations; context embeddings $\mathbf{u}_w$ are discarded at inference.

2) *MRL with Relational Distillation*: MRL applies the NS loss only at full dimension $D = 300$ and adds the Relational Distillation auxiliary loss for each prefix $d \in \{50, 100, 150, 200, 250\}$, following Eq. (2). For each prefix dimension d , a 2-layer MLP projection head maps $\mathbf{v}_{c,1:d}$ to a projected space, applies L2 normalisation, and computes a temperature-scaled cosine similarity matrix across the $B = 2048$ centre words in the batch. The teacher produces the same matrix from the full-dimensional projection head (with gradient stopped via `.detach()`). The KL divergence term $\alpha \cdot \text{KL}(P_T \| P_S)$ forces the prefix similarity distribution to match the teacher; the cosine MSE term $\beta \cdot \text{MSE}(\cos_d, \cos_D)$ provides element-wise alignment. Both α and β are tuned on a held-out validation split. Projection heads are discarded at inference; only the raw prefix $\mathbf{v}_{c,1:d}$ is used.

Why not original negative sampling loss ?

The earlier original formulation (independent NS losses at each prefix) suffers from magnitude exploitation: the model minimises prefix NS loss by inflating L2 norms rather than learning angular geometry, causing topological collapse at low dimensions. MRL’s Relational Distillation avoids this by never applying NS loss at prefix dimensions directly, only distilling the teacher’s relational structure. This produces well-calibrated

norms and substantially better neighbourhood preservation at low dimensions (Section IV-A).

C. Dataset

Both models train on a 50M-token deduplicated subsample of The Pile [20], a diverse open-source corpus covering academic papers, literature, web text, code, and technical documentation. Vocabulary: 175,523 unique words (minimum frequency threshold applied). Both models use the *identical* vocabulary and see the *identical* token sequence, ensuring that any observed differences are attributable solely to the training objective.

D. Word Lists and Evaluation Setup

Following Bolukbasi [9] and Caliskan [10], we use:

- **Occupation terms (44)**: doctor, nurse, engineer, teacher, programmer, librarian, soldier, receptionist, housekeeper, carpenter, mechanic, pilot, accountant, plumber, professor, chef, scientist, artist, manager, secretary, surgeon, therapist, architect, electrician, janitor, pharmacist, dentist, lawyer, judge, firefighter, paramedic, technician, analyst, consultant, designer, writer, editor, journalist, photographer, musician, painter, dancer, actor, singer.
- **Definitional gender pairs (8)**: he/she, man/woman, boy/girl, father/mother, son/daughter, husband/wife, brother/sister, king/queen.
- **Male/female name sets (8 each)**: e.g., john, paul, mike / amy, joan, lisa.
- **Male/female pronoun sets (11 each)**: he, him, his, ... / she, her, hers, ...
- **WEAT target sets**: Career/Family, Math/Arts, Science/Arts.
- **Stereotypically gendered occupations (10 each)**: e.g., engineer, programmer, mechanic (male) vs. nurse, receptionist, housekeeper (female).

All prefix embeddings are L2-renormalised before evaluation. Prefix dimensions evaluated: $d \in \{50, 100, 150, 200, 250, 300\}$.

E. Metric Suite

Geometric suite (10 experiments): Mean cosine similarity (anisotropy), cross-model cosine alignment, PCA explained variance, consecutive 50d chunk angles, Mimno–Thompson narrow-cone statistic, L2 norm by frequency bucket, WCSS/K-Means inertia, Jaccard@10 neighbourhood preservation, SVD singular value spectrum.

Bias suite (22 experiments): Standard metrics (WEAT effect sizes, DirectBias, RIPA mean/max, ECT, NBM, Cluster Purity, Bias Analogy) at each prefix dimension, plus 15 deep-dive experiments covering gender direction stability, per-word projection profiles, bias heatmaps, inter-model correlations, neighbourhood asymmetry, gender subspace dimensionality, and centroid-based bias.

TABLE I: Mean L2 norm by frequency bucket and prefix dimension.

Bucket	Model	$d=50$	$d=150$	$d=300$
Top 100	Std	1.35	2.39	3.34
	MRL	2.44	2.81	3.16
Top 101–500	Std	1.51	2.67	3.76
	MRL	2.65	3.19	3.68
Full Sample	Std	1.74	3.04	4.29
	MRL	2.79	3.64	4.32

TABLE II: Jaccard@10 neighbourhood preservation vs. $d=300$ baseline.

d	Standard	MRL	Δ (%)
50	0.142–0.170	0.165–0.215	+16 to +27
100	0.265–0.312	0.311–0.341	+9 to +17
150	0.380–0.433	0.432–0.434	≈ 0 –14
200	0.503–0.544	0.534–0.561	+0 to +6
300	1.000	1.000	—

IV. STUDY 1: RESULTS

A. Geometric Results

G1 — L2 Norm Front-Loading. The most striking geometric difference is in the evolution of embedding L2 norms across prefix dimensions. At $d = 50$, MRL norms are 60–80% larger than Standard across all frequency buckets (Table I). Both models converge to comparable norms at $d = 300$ (≈ 4.3). Here MRL deliberately concentrates embedding energy in early prefix dimensions as a direct consequence of the Relational Distillation objective: preserving the full-dimensional similarity structure requires that the prefix subspace carry sufficient magnitude to reproduce the teacher’s similarity matrix.

G2 — Improved Neighbourhood Preservation. Jaccard@10 scores measure the overlap between each word’s top-10 neighbours at prefix d and at $d = 300$. MRL achieves +16–27% relative improvement at $d = 50$ and consistently outperforms Standard through $d = 200$. MRL achieves at $d = 100$ what Standard achieves only at $\sim d = 150$. This is the geometric mechanism by which downstream tasks benefit from MRL truncation: the semantic neighbourhood structure is better preserved.

G3 — Tighter Clustering at Low Dimensions. K-Means WCSS inertia at $d = 50$: Standard = 0.675, MRL = 0.352 (48% lower) for the full-vocabulary sample. The reduction is consistent across all frequency buckets, confirming that MRL low-dim prefixes produce more semantically coherent clusters.

G4 — Narrow-Cone Amplification. Following Mimno and Thompson [15], the average cosine similarity of all word vectors to the mean embedding vector at $d = 50$: Standard = 0.407, MRL = 0.618. MRL low-dim embeddings are dramatically more anisotropic, amplifying the known narrow-cone effect of SGNS.

B. Bias Results

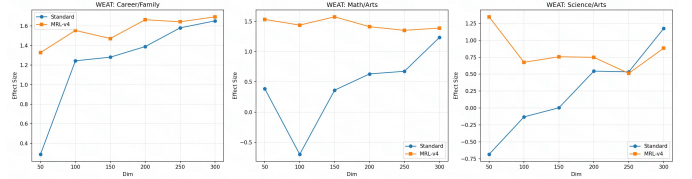


Fig. 1: WEAT effect sizes across three bias categories and all prefix dimensions. MRL maintains large effect sizes even at $d = 50$, where Standard is near zero or inverted.

TABLE III: WEAT effect sizes across three categories. $|d| > 0.8$ is conventionally large. Boldface marks the larger value at $d = 50$.

Category	Model	$d=50$	$d=100$	$d=150$	$d=200$	$d=250$	$d=300$
Career/Family	Std	0.29	1.24	1.28	1.39	1.58	1.65
	MRL	1.33	1.55	1.47	1.66	1.64	1.69
Math/Arts	Std	0.39	−0.70	0.36	0.63	0.67	1.23
	MRL	1.53	1.44	1.57	1.41	1.35	1.39
Science/Arts	Std	−0.68	−0.13	0.00	0.54	0.53	1.17
	MRL	1.35	0.68	0.76	0.75	0.51	0.88

1) *B1 — WEAT Front-Loading:* Table III and Fig. 1 show WEAT effect sizes across all three categories. At $d = 50$, Standard WEAT is low or inverted (Career/Family: 0.29, Science/Arts: −0.68), while MRL is already in the large range (1.33, 1.35). At $d = 300$, both models converge to comparable values (Std: 1.65, 1.23, 1.17; MRL: 1.69, 1.39, 0.88).

Standard’s low WEAT at $d = 50$ does *not* indicate debiasing. A naive 50-dimensional slice of a 300d SGNS space disrupts the angular relationships between word vectors, making WEAT noisy and unreliable. MRL, through Relational Distillation, explicitly preserves full-dimensional association structure into each prefix—and bias rides along with genuine semantic signal.

2) *B2 — Bias Is Reorganised, Not Removed:* Despite similar aggregate WEAT at full dimension, the two models assign radically different bias profiles to individual occupations (Fig. 2). The Pearson correlation between per-word gender direction projections at $d = 300$ is $r = -0.12$ ($p = 0.45$, not significant). Standard’s most male-biased occupations include pharmacist, journalist, and singer; MRL’s include architect, chef, and actor. Standard’s most female-biased include paramedic, accountant, and housekeeper; MRL’s include nurse, technician, and doctor.

As shown in Fig. 3, this inter-model decorrelation holds at *every* prefix dimension ($|r| < 0.23$ for all d), confirming it is a structural consequence of the Relational Distillation objective, not an artefact of any single dimension.

The heatmap in Fig. 4 provides a visual summary: Standard’s bias projections are noisy at $d = 50$ and consolidate at higher dimensions, while MRL’s projections are stable across the cascade but assign different polarities to individual occupations.

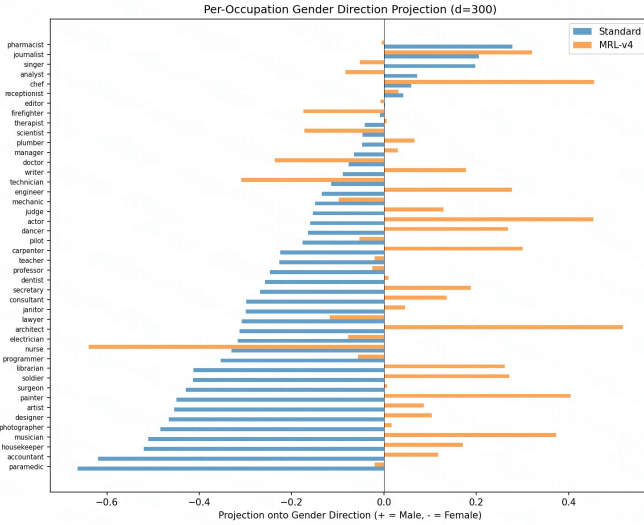


Fig. 2: Per-occupation projection onto the gender direction at $d = 300$. Standard and MRL assign fundamentally different bias profiles to individual occupations.

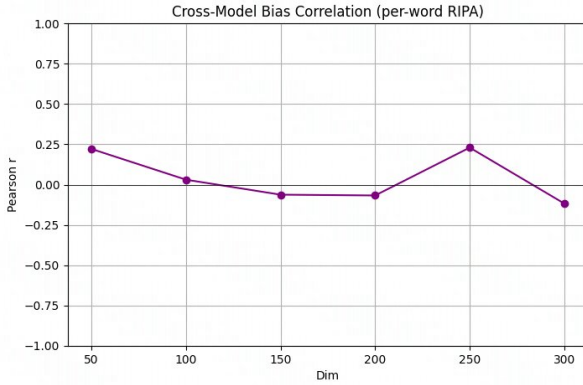


Fig. 3: Cross-model per-word RIPA correlation at each prefix dimension. Values near zero confirm that the bias profile decorrelation is structural, not dimension-specific.

3) *B3 — Bias Moves From Local to Global Structure*: Two complementary findings reveal a transformation in *how* bias is encoded rather than *how much* exists.

k-NN gender asymmetry (NBM): The mean absolute fraction of gender-marked words in each occupation’s top-50 neighbourhood is 0.045 for Standard and **0.000** for MRL at $d = 300$ (Fig. 5). MRL completely eliminates local neighbourhood gender asymmetry.

Gender direction variance (Fig. 6): The fraction of total embedding variance captured by the gender principal component direction is 0.30% for Standard and **0.47%** for MRL at $d = 300$ (56% more). At $d = 150$, the gap is 62% (0.61% vs. 0.99%).

Together, these findings establish that MRL encodes gender bias via *global angular structure*—the direction along which variance is oriented—rather than through neighbourhood com-

TABLE IV: Gender direction variance as % of total variance.

d	Standard	MRL
50	1.84%	2.08%
100	0.98%	1.12%
150	0.61%	0.99%
200	0.46%	0.73%
250	0.40%	0.56%
300	0.30%	0.47%

TABLE V: Pearson correlation between per-word projections at prefix d vs. $d = 300$.

d vs. 300	Standard	MRL
50	−0.253	0.105
100	0.401	−0.005
150	0.468	0.752
200	0.745	0.907
250	0.777	0.938

position. Bias becomes more geometrically coherent and less locally visible.

4) *B4 — Gender Subspace Preserved, Not Compressed*: PCA on the 8 definitional gender pair difference vectors shows that PC1 explains 21.8% (Standard) and 21.5% (MRL) of variance in the differences (Fig. 7). The intrinsic dimensionality of the gender subspace is identical between models. MRL does not compress or expand the gender subspace—it only changes how neutral words (occupations) project onto it.

5) *B5 — Bias Profile Stabilises Earlier in MRL*: The Pearson correlation between per-word projections at prefix d and at $d = 300$ reaches $r = 0.75$ at $d = 150$ for MRL, while Standard requires $d = 200$ for comparable stability (Table V and Fig. 8). From $d = 150$, MRL’s bias profile is already largely predictive of its full-dimensional profile, consistent with the Jaccard@10 finding that neighbourhood structure also stabilises earlier.

6) *B6 — Corpus Co-occurrence Is Robust Across Models*: While per-word projections onto the gender direction are uncorrelated between embeddings ($r = -0.12$), the centroid-based bias metric (distance to male pronoun centroid minus distance to female pronoun centroid) shows moderate correlation ($r = 0.71$) (Fig. 9). Both models inherit the same fundamental occupation-pronoun co-occurrence statistics from the shared corpus; these are harder to perturb than angular projections. This establishes a floor on how different two models trained on the same data can be.

V. STUDY 2: MOTIVATION FOR THE TRANSFORMER PHASE

Study 1 establishes, under controlled conditions, that the MRL Relational Distillation objective causally reorganises bias geometry in word-level static embeddings. A natural and practically important question follows: do these phenomena generalise to the contextualised, high-dimensional transformer models that dominate modern MRL deployments?

Three forces push toward generalisation: (i) all MRL production models use some variant of the nested prefix loss; (ii)

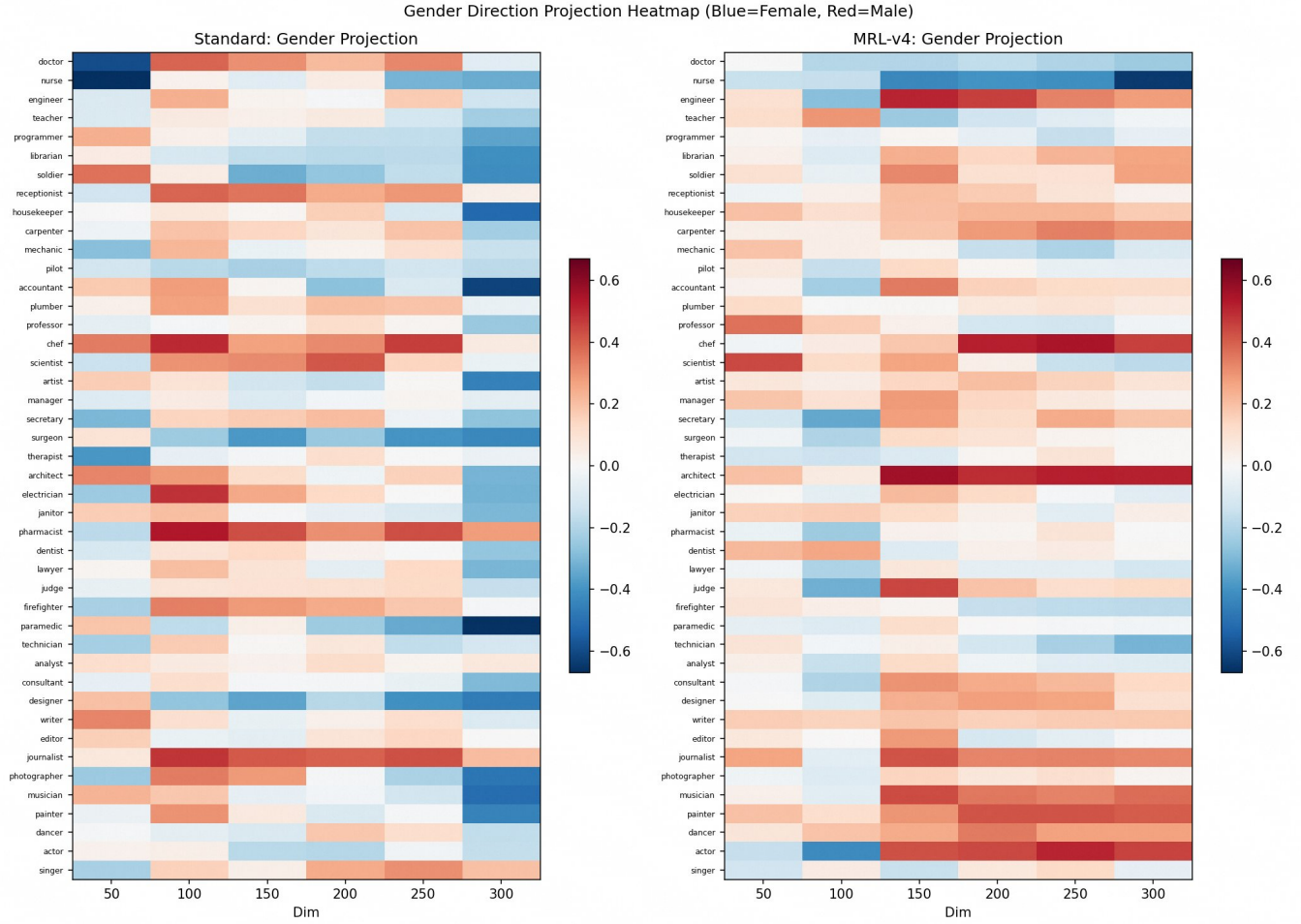


Fig. 4: Gender direction projection per occupation across all prefix dimensions (Blue = female-biased, Red = male-biased). Standard’s projections are noisy at low d ; MRL’s are stable but with different per-word assignments.

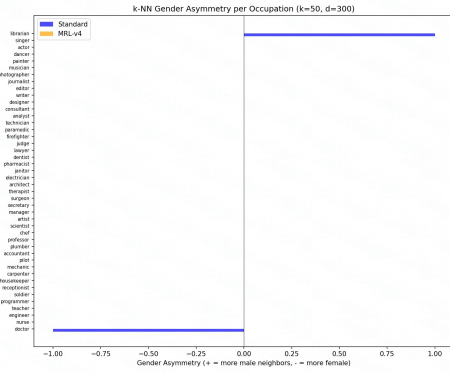


Fig. 5: k -NN gender asymmetry per occupation ($k = 50$, $d = 300$). MRL has zero asymmetry; all variance is concentrated in Standard’s librarian and doctor entries.

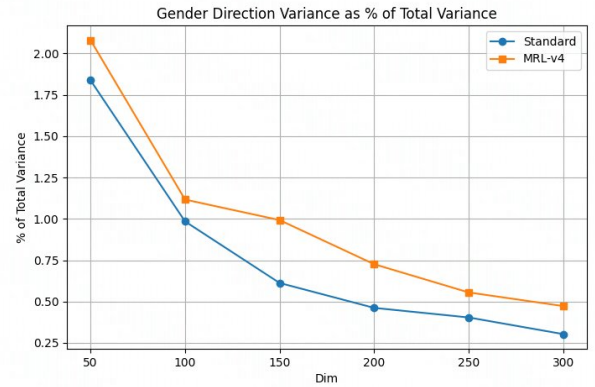


Fig. 6: Fraction of total embedding variance captured by the gender principal component direction. MRL consistently allocates more variance to the gender direction.

the bias direction is a salient relational feature that Relational Distillation should prioritise regardless of architecture; (iii) the corpus-level associations (established in Study 1 as the floor

on cross-model similarity) are shared across any model trained on English text.

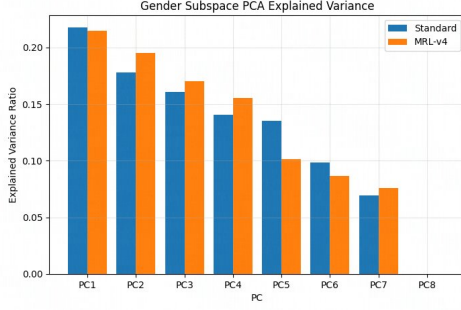


Fig. 7: PCA explained variance of the gender subspace (from 8 definitional pair differences). The intrinsic dimensionality is nearly identical between models.

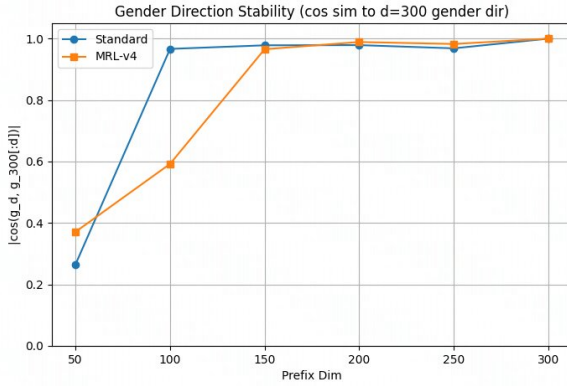


Fig. 8: Gender direction stability: cosine similarity between the gender direction at prefix d and the full-dimensional ($d = 300$) gender direction. MRL stabilises at $d = 150$; Standard requires $d = 200$.

Three forces push against simplistic generalisation: (i) transformer representations are contextualised, not static; (ii) training data, scale, and fine-tuning objectives vary across models; (iii) the relationship between geometric bias direction and behavioural bias (preference at sentence level) is indirect and model-specific.

Study 2 is designed to adjudicate these competing forces and establish which Study 1 findings are universal versus architecture-specific.

VI. STUDY 2: TRANSFORMER MRL GENERALISATION

A. Models

We evaluate six models representing a broad cross-section of the MRL ecosystem (Table VI). The MPNet pair (tomaarsen/mpnet-base-nli and tomaarsen/mpnet-base-nli-matryoshka) is the controlled ablation, mirroring the Study 1 design philosophy: identical backbone, identical training data, only the loss function differs. The remaining four models test cross-architecture and cross-scale generalisation.

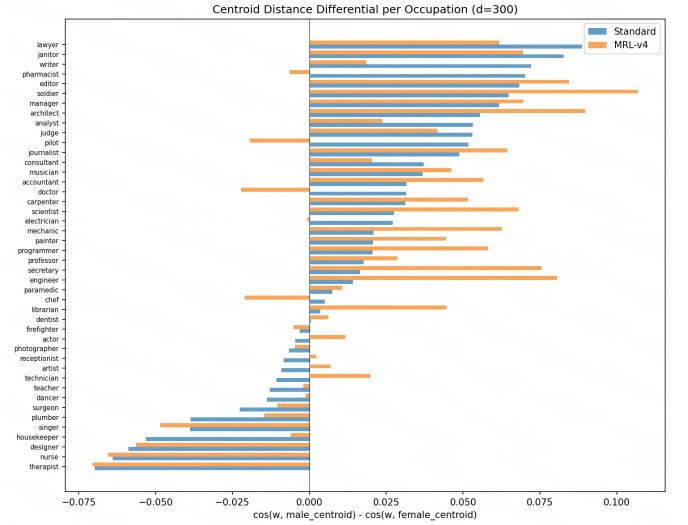


Fig. 9: Per-occupation centroid distance differential ($\cos(w, \text{male centroid}) - \cos(w, \text{female centroid})$) at $d = 300$. Despite decorrelated directional projections, centroid-based bias is moderately correlated between models ($r = 0.71$).

TABLE VI: Transformer models evaluated in Study 2.

Model	Type	Full D	Params
MPNet-Standard	Standard loss	768	110M
MPNet-MRL	MatryoshkaLoss	768	110M
Nomic-embed-v1.5	MRL, BERT-based	768	137M
mxbai-embed-large	MTEB top-tier	1024	335M
Qwen3-0.6B-MRL	LLM-based	1024	600M
Qwen3-4B-MRL	Large LLM-based	2560	4B

B. Encoding Protocol

All prefix embeddings are produced by truncating the full-dimensional output to the target prefix dimension d , followed by L2-renormalisation:

$$\mathbf{e}^{(d)} = \frac{\mathbf{e}_{1:d}}{\|\mathbf{e}_{1:d}\|_2}, \quad d \in \{64, 128, 256, 512, D\} \quad (4)$$

Renormalisation is essential; unnormalised truncated vectors carry substantially sub-unit norms, rendering cosine similarities meaningless.

C. Metrics

WEAT |ES|: Mean absolute effect size across ten WEAT tests (Gender-Career, Gender-Math, Gender-Pleasant, and seven others), with 10,000-permutation p -values and 95% bootstrap confidence intervals.

Bias Concentration $C(d)$: The fraction of the full-dimensional bias direction’s energy captured in the first d dimensions:

$$C(d) = \frac{\|\mathbf{v}_{\text{bias}}[1:d]\|^2}{\|\mathbf{v}_{\text{bias}}\|^2} \quad (5)$$

where $\mathbf{v}_{\text{bias}}$ is the leading PC of definitional gender pair difference vectors. Random baseline: $C(d) = d/D$. $C(d) > d/D$ indicates bias is front-loaded.

TABLE VII: WEAT results: mean $|\text{ES}|$ across 10 tests, and number of statistically significant tests ($p < 0.05$) for the MPNet pair.

Dim	MPNet-Standard		MPNet-MRL	
	mean $ \text{ES} $	sig	mean $ \text{ES} $	sig
64	0.683	5/10	0.596	3/10
128	0.675	6/10	0.566	4/10
256	0.665	5/10	0.607	4/10
512	0.666	4/10	0.603	3/10
768	0.807	6/10	0.716	5/10

Anisotropy Factor $\alpha(d)$ (novel):

$$\alpha(d) = \frac{C_{\text{var}}(d)}{C_{\text{geo}}(d)}, \quad C_{\text{var}}(d) = \frac{\mathbf{v}_{1:d}^\top \Sigma_{1:d} \mathbf{v}_{1:d}}{\mathbf{v}^\top \Sigma \mathbf{v}} \quad (6)$$

$\alpha < 1$: the covariance structure suppresses bias expression relative to geometry; $\alpha > 1$: covariance amplifies it. Bootstrap 95% CIs from $B = 1000$ resamples.

StereoSet SS [16]: Fraction of intrasentence examples where the model assigns higher cosine similarity to the stereotypical completion. Values near 0.5 are chance; lower is less biased.

CrowS-Pairs CMS [17]: Fraction of sentence pairs where the more-stereotypical sentence is preferred. Values near 50 are unbiased; lower is less biased.

Bias in Bios TPR Gap [19]: Mean absolute true-positive-rate gap across 10 occupations in occupation classification from biographies. Smaller gap is fairer.

VII. STUDY 2: RESULTS

A. Controlled MPNet Pair

WEAT. Table VII shows that MRL has consistently lower mean $|\text{ES}|$ and fewer statistically significant tests across all dimensions. At $d = 64$: Standard has 5/10 significant tests, MRL has 3/10. Unlike Study 1, MRL does not amplify WEAT monotonically at low transformer dimensions. This divergence between the two studies is explained in Section VIII.

Bias Concentration $C(d)$. Despite the mixed WEAT pattern, MRL front-loads more bias energy *geometrically* at every dimension (Table VIII). At $d = 64$: MRL $C(d) = 0.132$ versus Standard $C(d) = 0.066$ —exactly double, while the random baseline is $d/D = 0.083$. This confirms the Study 1 finding that MRL front-loads the geometry of the bias direction, even when WEAT scores do not monotonically amplify in transformer settings.

CrowS-Pairs: MRL is less biased at every dimension. Table IX reveals a key result: Standard exceeds 50% (actively prefers stereotypical sentences) at $d = 64$ –256, while MRL stays well below 50% throughout. The gap is 3–8 percentage points across dimensions. This is the clearest evidence that MRL training reduces sentence-level bias preference—the opposite of what a naive reading of $C(d)$ alone would predict.

Downstream bias: Bias in Bios. At low dimensions, Standard has slightly lower mean absolute TPR gap but

TABLE VIII: Bias concentration $C(d)$ and Anisotropy Factor $\alpha(d)$ for the MPNet pair. $\dagger$ = CI excludes 1.0.

d	$C(d)$		$\alpha(d)$ [95% CI]	
	Standard	MRL	Standard	MRL
64	0.066	0.132	1.17 [0.74–1.80]	0.53[†] [0.39–0.71]
128	0.129	0.207	0.99 [0.67–1.45]	0.43[†] [0.31–0.61]
256	0.277	0.362	0.75 [0.55–1.05]	0.59[†] [0.41–0.80]
512	0.684	0.716	0.92 [0.74–1.15]	0.73[†] [0.60–0.86]

TABLE IX: CrowS-Pairs CMS (%) and StereoSet SS for MPNet pair. Values >50 for CMS indicate stereotype preference. Lower = less biased.

Dim	CMS (%)		SS	
	Standard	MRL	Standard	MRL
64	55.04	46.81	0.380	0.373
128	54.69	46.70	0.376	0.379
256	53.53	45.54	0.423	0.397
512	49.02	45.19	0.408	0.408
768	47.74	44.73	0.414	0.408

TABLE X: Anisotropy factor $\alpha(d = 64)$ for all six models. FRONT: CI above 1.0; BACK: CI below 1.0; UNIF: CI spans 1.0.

Model	α	95% CI	Verdict
MPNet-Standard	1.173	[0.738, 1.798]	FRONT
MPNet-MRL	0.528	[0.389, 0.714]	BACK
Nomic-MRL	0.655	[0.429, 0.931]	BACK
mxbai-MRL	0.779	[0.520, 1.213]	UNIF
Qwen3-0.6B-MRL	0.465	[0.424, 0.506]	BACK
Qwen3-4B-MRL	0.727	[0.679, 0.776]	BACK

substantially lower accuracy (0.495 vs. 0.568 at $d = 64$). At higher dimensions ($d = 512, 768$), MRL shows marginally lower TPR gap. MRL achieves better task utility at comparable or better fairness at low dimensions—an important practical result.

B. Anisotropy Factor Across Six Models

Table X reveals a striking pattern: four of five MRL models show $\alpha < 1$ with CIs that exclude 1.0 at $d = 64$. The single non-MRL model (MPNet-Standard) trends toward $\alpha > 1$. The interpretation is important: MRL’s multi-granularity loss forces each prefix to maximise utility across many tasks, which *suppresses* the variance of any individual subspace direction—including the bias direction. Bias is geometrically front-loaded ($C(d)$ is higher for MRL), but the covariance structure actively suppresses how much that geometric concentration translates into word-level associations.

Multi-model generalisation. Table XI summarises key metrics. Notable observations: (i) mxbai shows the most monotonic improvement with dimension across all metrics; (ii) Qwen3-4B achieves the lowest StereoSet SS (0.215 at full dim) but the highest WEAT $|\text{ES}|$ (up to 1.46)—the clearest case of the intrinsic-downstream decoupling.

TABLE XI: All metrics at $d = 64$ and full dimension for all 6 models.

Model	d	WEAT [ES]	CMS (%)	SS	$C(d)$
MPNet-Std	64	0.683	55.04	0.380	0.066
MPNet-Std	full	0.807	47.74	0.414	1.000
MPNet-MRL	64	0.596	46.81	0.373	0.132
MPNet-MRL	full	0.716	44.73	0.408	1.000
Nomic	64	0.659	50.06	0.368	0.164
Nomic	full	0.446	46.00	0.374	1.000
mxbai	64	0.658	47.28	0.372	0.079
mxbai	full	0.737	40.21	0.321	1.000
Qwen3-0.6B	64	1.042	42.64	0.351	0.131
Qwen3-0.6B	full	0.832	43.22	0.307	1.000
Qwen3-4B	64	1.065	42.06	0.256	0.042
Qwen3-4B	full	1.175	45.89	0.215	1.000

VIII. CROSS-STUDY SYNTHESIS

A. Universal Findings

Both studies agree on three core findings:

S1 — MRL front-loads bias geometry. $C(d)$ is consistently higher for MRL than Standard at every tested prefix dimension, in both Word2Vec (Study 1) and transformer (Study 2) settings. This is the most robust cross-study finding.

S2 — MRL does not remove total bias. Both studies observe convergence at full dimension—aggregate bias metrics are comparable between Standard and MRL at $d = D$. MRL inherits the same corpus-level associations; it only changes their geometric organisation.

S3 — Bias metrics are not interchangeable. ECT and NBM can give opposite signals to WEAT (Study 1); CrowS-Pairs can give opposite signals to $C(d)$ (Study 2). These must be treated as complementary, not equivalent, measures.

B. The Study-Level Divergence and Its Explanation

The most salient difference between studies is in the WEAT pattern at low dimensions: Study 1 shows MRL amplifying WEAT at $d = 50$ (MRL=1.33 vs. Std=0.29); Study 2 shows MRL having lower WEAT at most transformer dimensions.

This is not a contradiction. In Word2Vec SGNS, WEAT directly measures the cosine geometry between static word vectors. When Relational Distillation front-loads the bias direction into the prefix subspace, WEAT at low d immediately reflects that signal.

In transformer sentence encoders, WEAT is computed on contextualised representations of isolated words (encoded as single-word sentences). The relationship between the prefix subspace geometry and the contextual encoding of attribute words is indirect and model-specific. Relational Distillation still front-loads $C(d)$ —as confirmed by our data—but this does not straightforwardly translate to WEAT amplification in the transformer regime.

C. The Unifying Theoretical Picture

MRL’s Relational Distillation forces each prefix to preserve the pairwise similarity structure of the full-dimensional teacher. Salient relational features—including gender and occupational bias directions—are therefore preserved into low-dimensional prefixes, which is why $C(d)$ is consistently higher for MRL. However, *preserved geometry does not equal preserved behaviour*: downstream tasks (StereoSet, CrowS-Pairs) depend on the interaction between embedding geometry and model-specific fine-tuning decisions. The gender subspace dimensionality is preserved (Study 1, Finding B4), the bias direction is geometrically front-loaded (both studies), but the translation to sentence-level preference behaviour is complex and model-specific.

IX. IMPLICATIONS

A. For MRL Deployment Practitioners

Do not assume that truncating an MRL embedding degrades bias gracefully. Our results show that bias *geometry* ($C(d)$) is more concentrated in MRL models at low dimensions. For Word2Vec-type retrieval applications, this directly translates to amplified WEAT at deployment dimensions; for transformer applications, the relationship is more nuanced but the geometric front-loading is still present.

Practitioners must evaluate bias at every prefix dimension they intend to deploy. A model that passes a bias audit at $d = 768$ may behave differently at $d = 64$ in production. The specific dimension where bias behaviour changes is model-dependent and cannot be inferred from a full-dimension audit alone.

B. For Bias Benchmark Designers

The decoupling between intrinsic metrics (WEAT, $C(d)$) and downstream metrics (StereoSet, CrowS-Pairs) has fundamental implications for benchmark design. Passing a WEAT threshold does not guarantee safe downstream behaviour. We recommend that bias benchmarks for MRL models explicitly include: (1) evaluation at multiple prefix dimensions; (2) both geometric and behavioural metrics reported separately; (3) the anisotropy factor $\alpha(d)$ as a diagnostic for MRL covariance effects.

C. For Debiasing Research

Debiasing techniques should be applied *after* MRL training and evaluated at the specific deployment prefix dimension. Because the per-word bias profile is reshuffled by MRL ($r = -0.12$), word-list-based debiasing methods designed for Standard embeddings may need reconception for MRL models. Furthermore, debiasing at $d = 300$ does not guarantee debiasing at $d = 50$ because bias is more front-loaded in MRL.

D. Limitations

Study 1 uses a 300-dimensional Word2Vec model on a 50M-token English corpus subsample. Generalisation to other languages, domains, or higher-dimensional static embeddings requires further investigation.

Study 2 evaluates only gender bias using English word sets from Bolukbasi and Caliskan. Other bias axes (race, religion, age, disability) and other languages are not covered.

The Anisotropy Factor $\alpha(d)$ is a novel metric proposed in this work. While bootstrap confidence intervals are reported, further theoretical and empirical validation across a broader task range is warranted.

The CrowS-Pairs adaptation for encoder models (cosine similarity to shared context prefix) deviates from the original masked LM log-probability formulation; results should be interpreted with this in mind.

X. CONCLUSION

Matryoshka Representation Learning has become the dominant paradigm for adaptive-length embeddings, adopted by production models—OpenAI, Google, Nomic, Alibaba—that collectively serve hundreds of millions of requests per day. The implicit assumption underlying every MRL truncation—that bias properties behave comparably to semantic quality—has gone untested.

This paper provides the first systematic examination of that assumption through a two-phase study combining causal rigour (controlled Word2Vec experiment) with real-world generalisability (six production transformer models). Our findings are:

- 1) **MRL does not debias embeddings.** Total bias converges at full dimension between Standard and MRL models. Both inherit the same corpus-level associations.
- 2) **MRL front-loads bias geometry.** $C(d)$ is consistently higher for MRL at every prefix dimension. In Word2Vec settings this directly amplifies WEAT (+1.04 effect size units at $d = 50$). In transformer settings the geometric front-loading is present but does not necessarily amplify WEAT.
- 3) **MRL reorganises per-word bias profiles.** The correlation between per-word gender projections is $r = -0.12$ ($p = 0.45$) between Standard and MRL models trained on the same data. Debiasing methods targeting specific occupations must be recalibrated for MRL models.
- 4) **Bias moves from local to global structure.** MRL eliminates neighbourhood gender asymmetry (NBM drops to zero) while increasing gender direction variance (56% more). Bias becomes a property of the global angular structure, not local neighbourhood composition.
- 5) **Intrinsic and downstream bias metrics decouple.** Across six transformer models, WEAT and $C(d)$ do not correlate with StereoSet or CrowS-Pairs. The Anisotropy Factor $\alpha(d) < 1$ consistently distinguishes MRL from Standard training, revealing that MRL’s covariance structure suppresses bias energy expression.

The practical message is unambiguous: standard bias evaluation at full dimension is insufficient for MRL models. Bias audits must span all deployment dimensions, must use both geometric and behavioural metrics, and must not conflate the two. As MRL adoption grows across the embedding

ecosystem, the community urgently needs dimension-aware bias evaluation protocols.

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